

Mahdi Ahmadi

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Professional Summary

Machine learning researcher and former Data Scientist with 4+ years of industry experience designing, evaluating, and productionizing ML systems. Delivered computer-vision, NLP, recommendation, safety, anomaly-detection, and automation projects with measurable operational and user impact. Experienced across experimentation, A/B testing, scalable data workflows, vector retrieval, technical leadership, and academic research.

Technical Skills

Programming and Data: Python, SQL, C, C++, Pandas, PySpark, MATLAB

Machine Learning: PyTorch, TensorFlow, scikit-learn, NLP, computer vision, recommendation systems, anomaly detection, causal inference, generative AI, probabilistic modeling

Production and Systems: Linux, Docker, Git, Airflow, Kubernetes, Grafana, OpenCV, FAISS, Milvus, Elasticsearch

Professional Experience

Graduate Research Associate, Arizona State University

Aug 2023 – Present

- Build and evaluate reproducible PyTorch and TensorFlow pipelines for computer vision, NLP, individual treatment-effect estimation, and counterfactual inference.
- Developed probabilistic methods for fair image generation, resulting in a SafeGenAI Workshop publication at NeurIPS 2024.
- Partner with the Arizona Department of Emergency and Military Affairs on predictive modeling for natural-disaster forecasting.
- Collaborate on experimental design, model evaluation, reproducibility, and technical documentation across academic and applied projects.

Graduate Teaching Assistant, Arizona State University

Aug 2024 – May 2025

- Supported CSE 571 (Artificial Intelligence) and CSE 471 (Introduction to AI) through office hours, grading, labs, and technical guidance.
- Helped students strengthen their understanding of search, reasoning, and practical AI problem solving.

Data Scientist, Divar

Oct 2019 – Jul 2023

- Designed, trained, shadow-tested, and deployed an automated advertisement acceptance system across 7 verticals, reaching 25% automation and saving work equivalent to 50 operations employees.
- Developed a fraud-detection model for the chat platform, improving precision by 8%.
- Built a harassment-detection model from text and numerical features, improving precision from 0.26 to 0.44.
- Developed an anomaly-detection model using text embeddings and time-series behavior to identify abnormal advertisements and support business decisions.
- Designed and A/B tested collaborative-filtering and content-based recommenders, increasing job-seeker engagement by 11%.
- Built production workflows in Python, PySpark, and Airflow and evaluated FAISS, Milvus, and Elasticsearch for large-scale recommendation and retrieval.
- Established weekly knowledge-sharing sessions, led data-science review meetings, and interviewed candidates as part of the technical hiring process.

Education

Arizona State University, Ph.D. in Computer Science Aug 2023 – Present
GPA: 4.00/4.00. Graduate coursework: Artificial Intelligence, Knowledge Representation, Probabilistic Reasoning and Learning, Mobile Computing.

Isfahan University of Technology, M.Sc. in Electrical and Electronics Engineering Sep 2016 – Apr 2019
Isfahan University of Technology, B.Sc. in Electrical and Electronics Engineering Sep 2012 – Apr 2016

Open-Source and Public Work

ReDMark github.com/MahdiShAhmadi/ReDMark
Open-source implementation of a deep-learning watermarking framework tied to published research.

Personal Website and Project Hub mahdishahmadi.github.io
Public portfolio of research, publications, code, and engineering background.

Honors and Service

Fulton Fellowship Award, Arizona State University 2023
Reviewer, IEEE Transactions on Multimedia and IEEE Transactions on Aerospace and Electronic Systems 2022
Second Place among M.Sc. Electrical Engineering Students, Isfahan University of Technology 2018

Selected Publications

Fair Image Generation from Pre-trained Models by Probabilistic Modeling. M. Ahmadi, J. Leland, A. Chatterjee, Y. Choi. SafeGenAI Workshop at NeurIPS, 2024.

ReDMark: Framework for Residual Diffusion Watermarking by Deep Networks. M. Ahmadi, A. Norouzi, N. Karimi, S. Samavi, A. Emami. Expert Systems with Applications, 2020.

A Learning-Inspired Strategy to Design Binary Sequences With Good Correlation Properties: SISO and MIMO Radar Systems. O. Rezaei, M. Ahmadi, M.M. Naghsh, A. Aubry, M.M. Nayebi, A. De Maio. IEEE Transactions on Aerospace and Electronic Systems, 2023.

Context-aware Saliency Detection for Image Retargeting Using Convolutional Neural Networks. M. Ahmadi, N. Karimi, S. Samavi. Multimedia Tools and Applications, 2021.

Lossless Compression of Angiogram Foreground with Visual Quality Preservation of Background. M. Ahmadi, A. Emami, M. Hajabdollahi, S.M.R. Soroushmehr, N. Karimi, S. Samavi, K. Najarian. IEEE EMBC, 2018.